Adaptive FIR System Identification Using LMS-Type Algorithms

Comparative MATLAB and Python Implementation of Standard LMS, Sign-LMS, and Sign-Sign LMS for Unknown System Modeling.

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Technologies:

MATLAB, Python, Adaptive Filtering, DSP, FIR Filters, System Identification

1. Project Overview

Adaptive system identification is a fundamental application of Digital Signal Processing (DSP), where an adaptive filter learns the behavior of an unknown system using only observed input-output data. Such techniques are widely used in communication systems, channel estimation, echo cancellation, adaptive control, and signal enhancement.

This project investigates three LMS-type adaptive algorithms:

- Standard LMS
- Sign-LMS
- Sign-Sign LMS

The objective is to evaluate their ability to identify an unknown linear time-invariant (LTI) system modeled as a second-order Butterworth low-pass filter. Particular attention is given to convergence behavior, mean squared error (MSE), coefficient adaptation, and frequency-domain accuracy.

2. Problem Statement

Many communications and embedded DSP systems require adaptive modeling of unknown channels or dynamic systems. Traditional optimization techniques are often computationally expensive for real-time applications.

LMS-type algorithms provide a practical alternative by iteratively updating filter coefficients using error feedback. Sign-based variants further reduce computational complexity by minimizing multiplier operations, making them attractive for embedded and resource-constrained systems.

The goal of this project is to compare the performance and computational trade-offs of Standard LMS, Sign-LMS, and Sign-Sign LMS algorithms for adaptive FIR system identification.

3. System Model

The unknown system is represented by a second-order Butterworth low-pass filter.

A white Gaussian input signal is applied simultaneously to:

1. The unknown system.
2. The adaptive FIR filters.

The adaptive filter output is compared with the desired system response to generate an error signal.

The error signal drives coefficient adaptation according to the selected LMS-type algorithm.

System Identification Architecture:

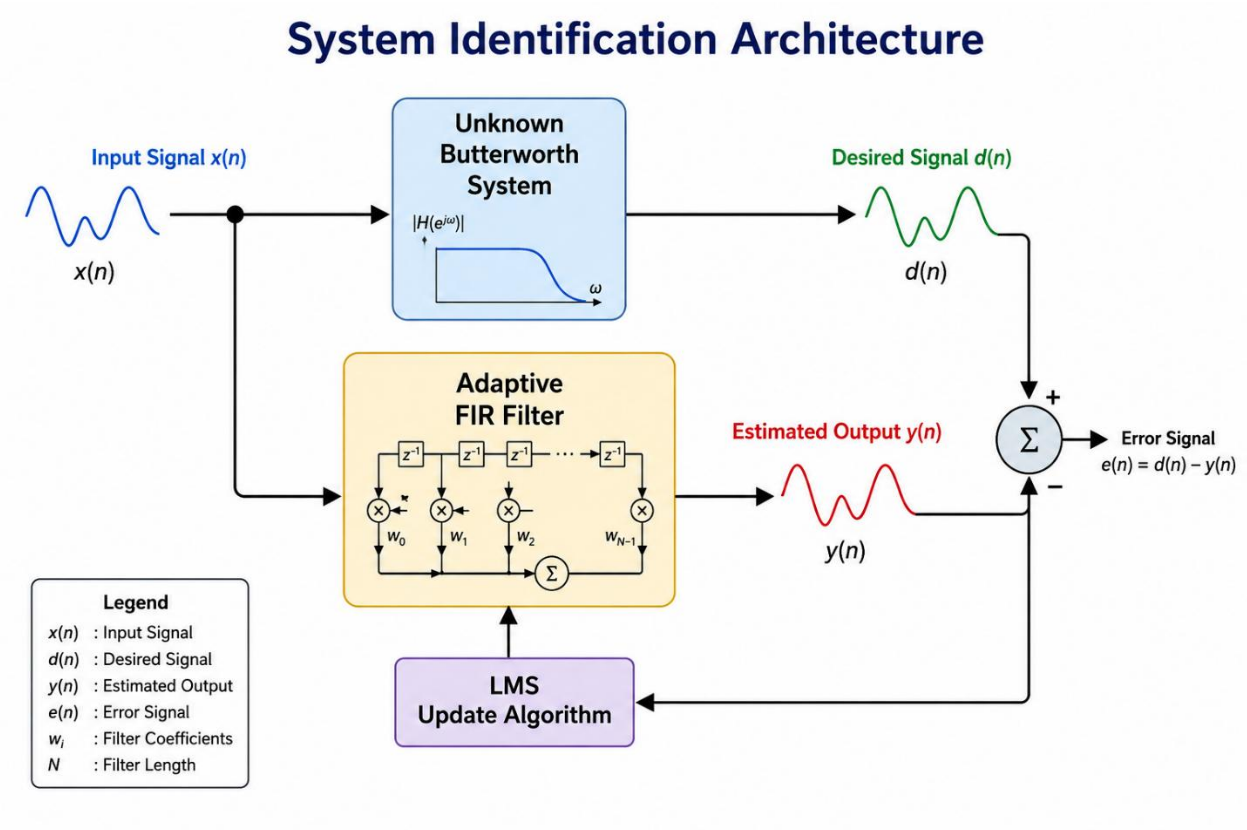

Figure 1: Block Diagram of LMS-Based Adaptive System Identification

The objective is to minimize the Mean Squared Error (MSE):

$$J = E[e^2(n)]$$

4. Algorithm Implementation

Standard LMS

The Standard LMS algorithm updates filter coefficients using the instantaneous error and input vector:

$$w(n+1) = w(n) + \mu e(n)x(n)$$

Advantages:

- Fast convergence
- High modeling accuracy

- Good steady-state performance

Limitations:

- Requires multipliers
- Higher computational cost

Sign-LMS

The Sign-LMS algorithm replaces the error term with its sign value:

$$w(n+1) = w(n) + \mu \, sign\big(e(n)\big) \, x(n)$$

Advantages:

- Reduced computational complexity
- Improved robustness to impulsive disturbances

Limitations:

- Slower convergence
- Higher residual error

Sign-Sign LMS

The Sign-Sign LMS algorithm replaces both the error and input terms with sign values:

$$w(n+1) = w(n) + \mu \, sign\big(e(n)\big) \, sign\big(x(n)\big)$$

Advantages:

- Very low computational cost
- Suitable for fixed-point implementations

Limitations:

- Slowest convergence
- Largest steady-state error

5. Results and Analysis

The algorithms were evaluated using:

- Mean Squared Error (MSE)
- Convergence Rate
- Coefficient Evolution
- Frequency Response Accuracy

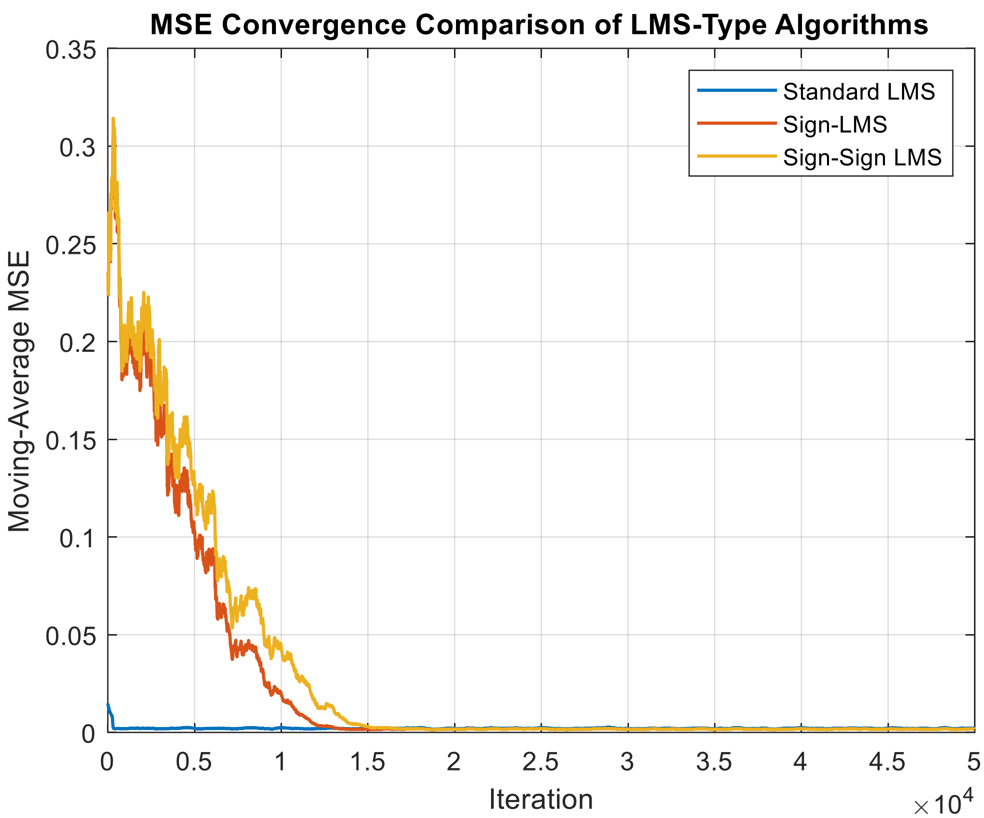

Figure 2: MSE Convergence Comparison of Standard LMS, Sign-LMS, and Sign-Sign LMS

Figure 2 compares the convergence performance of the three adaptive algorithms during system identification. The Standard LMS algorithm converges the fastest and achieves the lowest steady-state MSE. Sign-LMS and Sign-Sign LMS require more iterations to converge due to their simplified update rules but eventually reach stable solutions. These results demonstrate the trade-off between convergence speed and computational complexity.

Key Takeaway

Standard LMS provides the fastest convergence and highest identification accuracy, while the sign-based algorithms offer lower computational cost for embedded DSP applications.

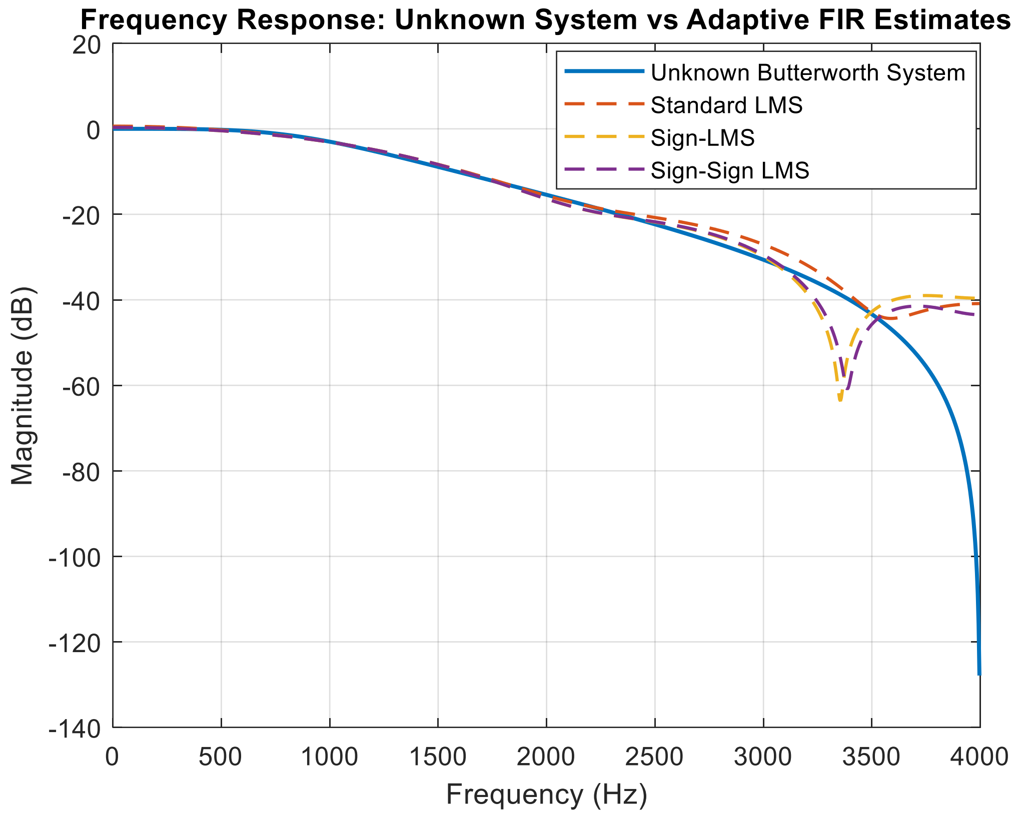

Figure 3: Frequency Response Comparison Between the Unknown System and Adaptive FIR Estimates

Figure 3 compares the estimated frequency responses of the adaptive FIR filters with the unknown Butterworth system. All three algorithms successfully approximate the system response across most of the frequency range. Small deviations are observed in the stopband due to the finite FIR filter length and differences in the adaptation process. Overall, Standard LMS provides the closest approximation to the reference system.

Key Takeaway

All algorithms successfully identify the unknown system, with Standard LMS producing the most accurate frequency response.

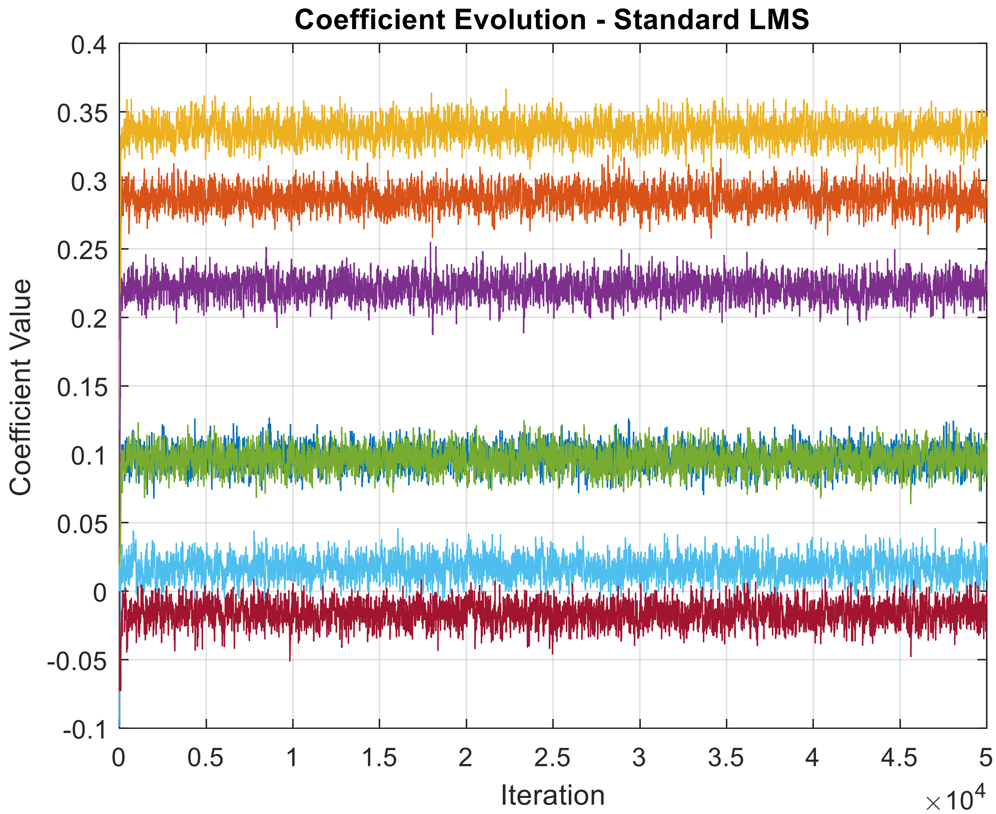

Figure 4: Coefficient Evolution of the Standard LMS Algorithm

Figure 4 illustrates the evolution of the adaptive FIR coefficients using the Standard LMS algorithm. The coefficients quickly converge from their initial values and stabilize with only minor fluctuations caused by measurement noise. This behavior confirms the stability and fast convergence characteristics of the Standard LMS algorithm.

Key Takeaway

The Standard LMS algorithm rapidly converges to stable filter coefficients, enabling accurate system identification.

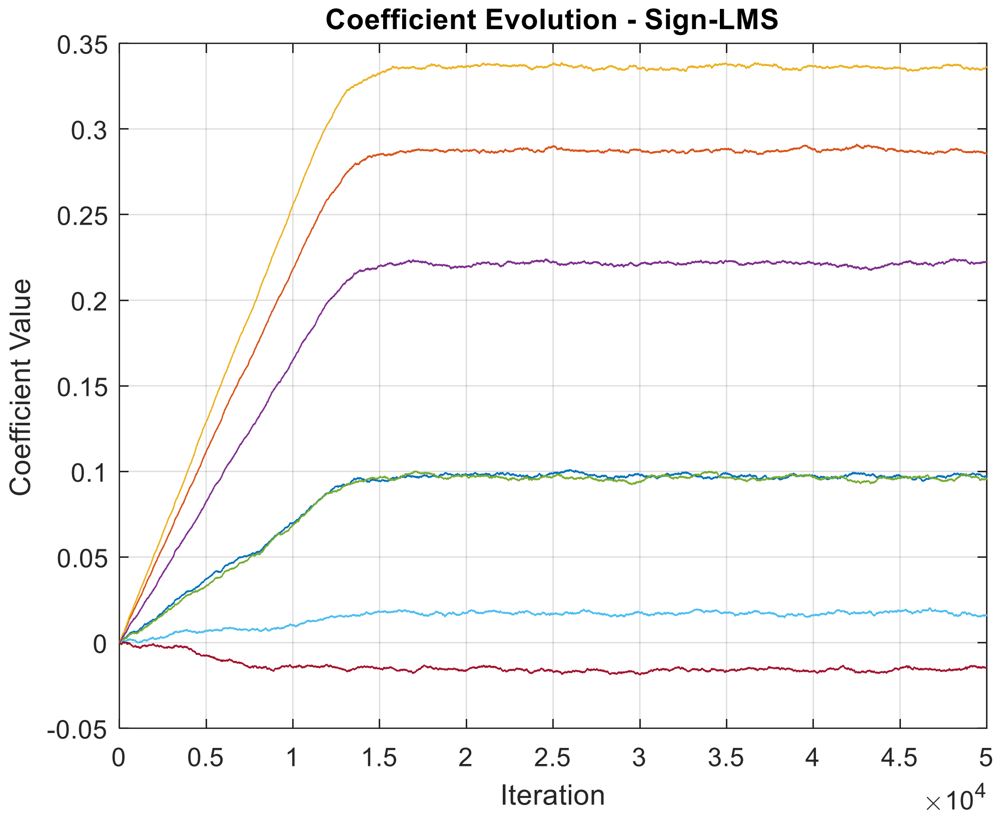

Figure 5: Coefficient Evolution of the Sign-LMS Algorithm

Figure 5 shows the coefficient adaptation process for the Sign-LMS algorithm. Compared to Standard LMS, the coefficients converge more gradually before reaching stable values. The slower convergence results from the simplified sign-based update mechanism, which reduces computational complexity while maintaining reliable estimation performance.

Key Takeaway

Sign-LMS achieves stable system identification with lower computational complexity, making it suitable for embedded implementations.

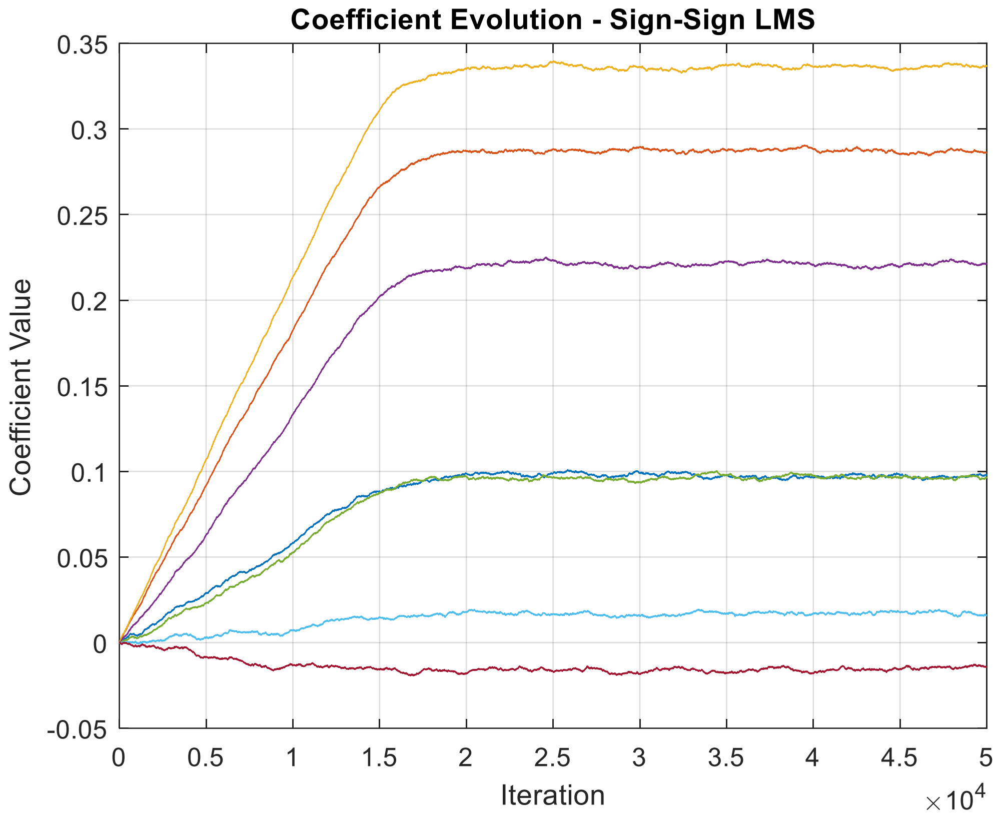

Figure 6: Coefficient Evolution of the Sign-Sign LMS Algorithm

Figure 6 presents the coefficient evolution using the Sign-Sign LMS algorithm. The coefficients converge more slowly than the other two algorithms but eventually stabilize around their final values. This behavior reflects the algorithm's highly simplified update rule, which significantly reduces arithmetic operations while preserving acceptable identification performance.

Key Takeaway

Sign-Sign LMS provides the lowest computational complexity, making it well suited for resource-constrained embedded DSP systems.

Key Observations

Standard LMS consistently achieved the fastest convergence and lowest steady-state error.

Sign-LMS reduced computational requirements while maintaining acceptable identification performance.

Sign-Sign LMS provided the lowest arithmetic complexity but exhibited slower adaptation and higher residual error.

Frequency response analysis showed that Standard LMS produced the closest approximation to the unknown Butterworth system.

6. Computational Trade-Off Analysis

Algorithm	Convergence Speed	Computational Cost	Accuracy
Standard LMS	High	Moderate	High
Sign-LMS	Moderate	Low	Moderate
Sign-Sign LMS	Low	Very Low	Lower

The results demonstrate the classic DSP engineering trade-off between computational complexity and adaptive performance.

When accuracy and convergence speed are critical, Standard LMS is preferred. For embedded platforms with strict power or hardware limitations, Sign-LMS and Sign-Sign LMS provide attractive alternatives.

8. Conclusion

This project demonstrates the effectiveness of LMS-type adaptive algorithms for FIR system identification. Standard LMS achieved the best overall performance in terms of convergence speed and modeling accuracy, while Sign-LMS and Sign-Sign LMS provided reduced computational complexity suitable for embedded implementations.

The study highlights how adaptive filtering algorithms can be selected according to system requirements, balancing accuracy, convergence behavior, computational cost, and hardware constraints.